

Nikita Jayaprakash

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EDUCATION

Ph.D. in Demography University of California, Berkeley	2026 - PRESENT
<i>Interests: infectious diseases, social networks, statistical techniques</i>	
Masters of Science in Applied Data Science Carnegie Mellon University	2025 - 2026
Bachelor's of Arts in Statistics University of California, Berkeley	2022 - 2025
<i>Cluster in Mathematics, Minors in Demography and English</i>	
Pembroke Cambridge Summer Programme University of Cambridge	Summer 2023

SKILLS & QUALIFICATIONS

- **Programming Languages:**
 - Python (pandas, numpy, matplotlib, xgboost, scikit-learn, pytorch, streamlit)
 - R (dplyr, tidyr, lubridate, ggplot2, caret, xgboost, randomForest, purrr, Shiny)
- **Additional Tools:** SQL, GitHub, LaTeX, Microsoft Suite (Office, Excel, Powerpoint)
- **Technical Skills:** practical applications of statistics and AI, data-driven best practices, machine learning, analytical problem-solving
- **Soft Skills:** interpersonal skills, written and oral communication skills, public speaking, organized, leadership skills, creativity
- **Domain Experience:** academic research, education, academic writing (IMRad), open-source software, nonprofit environments

ACADEMIC EXPERIENCE

Graduate Student Researcher	Aug 2026 - PRESENT
<i>Department of Demography, UC Berkeley</i>	
• Conduct research broadly related to the dynamics of infectious diseases under the direction of a faculty member	
Senior AI Curriculum Fellow	July 2026 - PRESENT
<i>D-Lab, UC Berkeley</i>	
• Facilitated AI literacy workshops and supported D-Lab programming for the broader Berkeley research community	
• Developed new workshop curriculum to teach AI-assisted programming techniques for a diverse audience	
Research Assistant	June 2026 - Aug 2026
<i>NSF AI Institute for Societal Decision Making, Carnegie Mellon University</i>	
• Developed products for emergency response in a partnership between CMU's AI-SDM and Meta's AI for Good program	
• Produced real-time dynamic situation reports aimed at helping first responders better address natural disasters like earthquakes, hurricanes and severe winter storms, utilizing Meta's limited-access mobility datasets	
• Explored the applications of computer vision techniques to provide actionable insights for on-the-ground usage during a disaster	
AI Fellow	Aug 2025 - May 2026
<i>NSF AI Institute for Societal Decision Making, Carnegie Mellon University</i>	
• Received inaugural NSF AI-SDM Fellowships for CMU master's of applied data science students	
• Presented at and attended workshops for 50+ hybrid participants (for industry and general public) on generative AI Literacy	
Research Assistant	Aug 2025 - May 2026
<i>Department of Statistics, UC Berkeley</i>	
• Assisted in experiment that compares grading accuracy across experts (faculty), student graders and multimodal LLMs (AI)	
• Developed workflow to streamline study design while maintaining student privacy (CITI program: Social & Behavioral Research)	
• Developed Python scripts that remove identifiable data and R package to compare the accuracy across the grading methods	
Data Science Student Researcher	Sept 2025 - Dec 2024
<i>Aspiranet</i>	
• Presented in collaboration with 5 students about therapy attendance of foster youth at UC Berkeley's Discovery Symposium	
• Utilized data analytics and statistical software to identify patterns in cancellation behavior over 3 years of data of 2000+ clients	
• Advised representative on effectiveness of text message reminders in reducing therapy appointment cancellations among youth	
Student Researcher	Feb 2023 - May 2025
<i>Department of Statistics, UC Berkeley</i>	
• Co-founded <i>NemoGB</i> , an instructor-facing learning management system deployed on main Berkeley software repository	
• Received \$34,000 grant from Statistics Department, impacting ~10,000 students across large data science and statistics courses	
• Presented live demos to professors in order to demonstrate grading functionality and receive user feedback during development	
• Leveraged the principles of data analysis and software development in order to make grading an informative experience	

PUBLICATIONS

- **Jayaprakash, Nikita**, Voutov, Monika, Turner, Zach, and Andrew Bray. NemoGB: An R Package for Calculating Course Grades. *Journal of Open Source Education*, 8(94), 303. <https://doi.org/10.21105/jose.00303>.

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PROFESSIONAL EXPERIENCE

Development Research Intern

Jan 2025 - June 2025

Major Giving Department, KQED

San Francisco, CA

- Conducted qualitative research of 100+ donor profiles and strategic event briefings for Development and Executive Management
- Developed an interactive 15-tab Excel spreadsheet of 1200+ donors for cross-portfolio comparison to guide business strategy

Data, Audits & Contracts Intern

June 2024 - Aug 2024

Education & Employment Services Department, Fortune Society

Queen, NY

- Audited data about reported demographics, job placements and client assessments in order to assess contract compliance
- Analyzed 1450+ formerly incarcerated clients' data (e.g. demographics, career milestones) over the past two years that resulted in a focused effort to address areas of outreach to increase program retention

AI / Data Science / ML / Developer Intern

Jan 2024 - May 2024

Hyphenova

Remote

- Led exploratory data analysis team in guiding decision-making with quantitative analysis of 700+ social media creators
- Built a Shiny app (in Python) to allow for quick analysis of creators in order to assess accuracy of statistical modeling

TEACHING EXPERIENCE

High School Instructor

June 2026 - July 2026

Academic Talent Development Program, UC Berkeley

Berkeley, CA

- Taught two 6-week summer courses (Applied Statistics in R; Data Science with R) to advanced high schoolers
- Developed the entire curriculum for Applied Statistics in R to introduce statistics through the lens of experimental design

High School Instructor

June 2025 - July 2025

Academic Talent Development Program, UC Berkeley

Berkeley, CA

- Taught 6-week summer course to 16 advanced high schoolers via UC Berkeley's Academic Talent Development Program
- Modified curriculum to include a final project using R Shiny while maintaining STEM course and program objectives
- Employed flipped-classroom learning environment to maximize hands-on coding experience with real-world datasets

Stat 20 Group Tutor

Jan 2023 - May 2025

Department of Statistics, UC Berkeley

Berkeley, CA

- Appointed to help students via Statistics Scholar program (intended to assist students of diverse backgrounds within the course) by communicating with students, planning group tutoring sessions and conducting quiz reviews
- Supported 100+ students and coordinated with professor in interactive lectures and assignments in an introductory statistics course

ACADEMIC PROJECTS

Vooks Behavioral Signals in User Engagement Capstone Project

Spring 2026

- Predicted trial conversion and subscription churn with machine learning models (LASSO logistic regression, XGBoost, balanced random forest) and survival analysis techniques (time-varying cox proportional hazard model) for a storybook streaming platform
- Presented actionable business insights based on stakeholder input, data-driven recommendations, and qualitative industry research

Deep Learning and Clustering Analysis of Gene Expression Project

Spring 2026

- Utilized applications of deep learning (variable auto-encoders), statistical genomics, and high-dimensional inference
- Partition the 3005 cells into biologically meaningful groups that align with known cell types and generated low-dimensional embeddings that provide insight into the underlying structure and separability of the single-cell RNA sequencing of mice

U.S. Weekly Gasoline Prices Time Series Project

Spring 2026

- Utilized vector autoregressive methods in order to analyze the relationship of American gasoline prices with crude oil prices and comparing regional gasoline prices within the United States in order to model, forecast and understand US gasoline price dynamics

ETL (Extract, Transform, Load) Data Pipeline Project

Fall 2025

- Designed database structure and table schema for Postgres SQL database design with a team of 2 other graduate students using large data sets from Health and Human Services and hospital quality information from Centers for Medicare and Medicaid Services
- Develop reports and data analytics using Python's Streamlit to summarize data with benchmarking

Goodreads Textual Analysis Project

Fall 2025

- Developed a web scraper to scrape 2713 Goodreads descriptions of books banned in the United States from 2023 to 2024
- Wrote a full IDMRaD report that used multi-dimensional factor analysis of the Biber features and keyness in order to determine literary features and literary differences between the Goodreads descriptions of more-banned and less-banned books

HIV Treatment Data Analysis Project

Fall 2025

- Utilized linear regression models to analyze 2139 observations from a randomized clinical trial for HIV treatment data
- Investigated prolonged clinical benefit from drug intervention and responses to HIV therapy according to progression of disease and prior HIV drug exposure using observational causal inference methodologies